

Model 1:
Keras1 layer Json
LS:

Overview

Name	Latency	Minimum
▼ myproject	7	7
dense_latency<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, config2>	1	1
relu<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, relu_config3>	0	0
dense_latency<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, config4>	1	1
sigmoid<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, sigmoid_config5>	2	2

Vitis:

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== Vitis HLS Report for 'myproject'
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* Date:      Sat Aug 29 13:31:58 2026

* Version:    2025.2.1 (Build 6397637 on Mar 18 2026)
* Project:    myproject_prj
* Solution:    solution1 (Vivado IP Flow Target)
* Product family: virtexuplus
* Target device: xcvu13p-flga2577-2-e
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== Performance Estimates
=====
+ Timing:
  * Summary:
    +-----+-----+-----+-----+
    | Clock | Target | Estimated| Uncertainty|
    +-----+-----+-----+-----+
    |ap_clk | 5.00 ns| 3.490 ns|   1.35 ns|
    +-----+-----+-----+-----+

+ Latency:
  * Summary:
    +-----+-----+-----+-----+-----+-----+-----+
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Latency (cycles)		Latency (absolute)		Interval		Pipeline	
min	max	min	max	min	max	Type	
+-----+-----+-----+-----+-----+-----+							
7	7	35.000 ns	35.000 ns	1	1	yes	
+-----+-----+-----+-----+-----+-----+							

+ Detail:
 * Instance:

+-----+-----+-----+-----+-----+-----+							
-----+-----+-----+-----+-----+-----+							
Latency (cycles)		Latency (absolute)		Interval		Pipeline	
		Instance				Module	
min	max	min	max	min	max	Type	
+-----+-----+-----+-----+-----+-----+							
-----+-----+-----+-----+-----+-----+							
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s_fu_49							
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s				1		1	
5.000 ns		5.000 ns		1 1		yes	
call_ret2_relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s_fu_55							
relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s				0		0	
0 ns		1 1		yes		0 ns	
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91							
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s				1		1	
5.000 ns		5.000 ns		1 1		yes	
grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127							
sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s				2		2	
10.000 ns		10.000 ns		1 1		yes	
+-----+-----+-----+-----+-----+-----+							
-----+-----+-----+-----+-----+-----+							

* Loop:
 N/A

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 == Utilization Estimates
 =====
 * Summary:
 +-----+-----+-----+-----+-----+-----+

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	10	-
FIFO	-	-	-	-	-
Instance	1	305	1041	10551	-
Memory	-	-	-	-	-
Multiplexer	-	-	0	36	-
Register	-	-	1212	-	-
Total	1	305	2253	10597	0
Available SLR	1344	3072	864000	432000	320
Utilization SLR (%)	~0	9	~0	2	0
Available	5376	12288	3456000	1728000	1280
Utilization (%)	~0	2	~0	~0	0

+ Detail:

* Instance:

+-----+-----+-----+-----+-----+-----+								
		Instance				Module		
BRAM_18K		DSP		FF		LUT		URAM
+-----+-----+-----+-----+-----+-----+								
-----+-----+-----+-----+-----+-----+								
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s_fu_49								
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s				0		275		850
8425		0						
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91								
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s				0		30		178
797		0						
call_ret2_relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s_fu_55								
relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s				0		0		0 1216
0								
grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127								
sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s				1		0		13
113		0						

	Total				
	1	305	1041	10551	0

* DSP:

N/A

* Memory:

N/A

* FIFO:

N/A

* Expression:

Variable Name		Operation	DSP	FF	LUT
			Bitwidth P0	Bitwidth P1	
ap_block_pp0_stage0_11001		and	0 0 2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp10		and	0 0 2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp77		and	0 0 2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp79		and	0 0 2	1	1
ap_enable_pp0		xor	0 0 2	1	2
Total			0 0 10	5	6

* Multiplexer:

Name		LUT	Input Size	Bits
				Total Bits
input_1_ap_vld_in_sig		9	2	1
input_1_ap_vld_preg		9	2	1
input_1_blk_n		9	2	1
input_1_in_sig		9	2	160
Total		36	8	163

* Register:

	Name	FF	LUT	Bits	Const Bits
	ap_CS_fsm	1	0	1	0
0	ap_block_pp0_stage0_subdone_grp0_done_reg		1	0	1
	ap_enable_reg_pp0_iter1	1	0	1	0
	ap_enable_reg_pp0_iter2	1	0	1	0
	ap_enable_reg_pp0_iter3	1	0	1	0
	ap_enable_reg_pp0_iter4	1	0	1	0
	ap_enable_reg_pp0_iter5	1	0	1	0
	ap_enable_reg_pp0_iter6	1	0	1	0
	ap_enable_reg_pp0_iter7	1	0	1	0
	grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91_ap_start_reg	1	0	1	0
	grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127_ap_start_reg	1	0	1	0
	input_1_ap_vld_preg	1	0	1	0
	input_1_preg	160	0	160	0
	layer2_out_10_reg_445	16	0	16	0
	layer2_out_11_reg_450	16	0	16	0
	layer2_out_12_reg_455	16	0	16	0
	layer2_out_13_reg_460	16	0	16	0
	layer2_out_14_reg_465	16	0	16	0
	layer2_out_15_reg_470	16	0	16	0
	layer2_out_16_reg_475	16	0	16	0
	layer2_out_17_reg_480	16	0	16	0
	layer2_out_18_reg_485	16	0	16	0
	layer2_out_19_reg_490	16	0	16	0
	layer2_out_1_reg_400	16	0	16	0
	layer2_out_20_reg_495	16	0	16	0
	layer2_out_21_reg_500	16	0	16	0
	layer2_out_22_reg_505	16	0	16	0
	layer2_out_23_reg_510	16	0	16	0
	layer2_out_24_reg_515	16	0	16	0
	layer2_out_25_reg_520	16	0	16	0
	layer2_out_26_reg_525	16	0	16	0
	layer2_out_27_reg_530	16	0	16	0
	layer2_out_28_reg_535	16	0	16	0
	layer2_out_29_reg_540	16	0	16	0

layer2_out_2_reg_405	16 0 16	0
layer2_out_30_reg_545	16 0 16	0
layer2_out_31_reg_550	16 0 16	0
layer2_out_3_reg_410	16 0 16	0
layer2_out_4_reg_415	16 0 16	0
layer2_out_5_reg_420	16 0 16	0
layer2_out_6_reg_425	16 0 16	0
layer2_out_7_reg_430	16 0 16	0
layer2_out_8_reg_435	16 0 16	0
layer2_out_9_reg_440	16 0 16	0
layer2_out_reg_395	16 0 16	0
layer3_out_10_reg_605	16 0 16	0
layer3_out_11_reg_610	16 0 16	0
layer3_out_12_reg_615	16 0 16	0
layer3_out_13_reg_620	16 0 16	0
layer3_out_14_reg_625	16 0 16	0
layer3_out_15_reg_630	16 0 16	0
layer3_out_16_reg_635	16 0 16	0
layer3_out_17_reg_640	16 0 16	0
layer3_out_18_reg_645	16 0 16	0
layer3_out_19_reg_650	16 0 16	0
layer3_out_1_reg_560	16 0 16	0
layer3_out_20_reg_655	16 0 16	0
layer3_out_21_reg_660	16 0 16	0
layer3_out_22_reg_665	16 0 16	0
layer3_out_23_reg_670	16 0 16	0
layer3_out_24_reg_675	16 0 16	0
layer3_out_25_reg_680	16 0 16	0
layer3_out_26_reg_685	16 0 16	0
layer3_out_27_reg_690	16 0 16	0
layer3_out_28_reg_695	16 0 16	0
layer3_out_29_reg_700	16 0 16	0
layer3_out_2_reg_565	16 0 16	0
layer3_out_30_reg_705	16 0 16	0
layer3_out_31_reg_710	16 0 16	0
layer3_out_3_reg_570	16 0 16	0
layer3_out_4_reg_575	16 0 16	0
layer3_out_5_reg_580	16 0 16	0
layer3_out_6_reg_585	16 0 16	0
layer3_out_7_reg_590	16 0 16	0
layer3_out_8_reg_595	16 0 16	0
layer3_out_9_reg_600	16 0 16	0
layer3_out_reg_555	16 0 16	0
layer4_out_reg_715	16 0 16	0

Total	1212 0 1212 0
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== Interface
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* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
ap_clk	in	1	ap_ctrl_hs	myproject	return value
ap_rst	in	1	ap_ctrl_hs	myproject	return value
ap_start	in	1	ap_ctrl_hs	myproject	return value
ap_done	out	1	ap_ctrl_hs	myproject	return value
ap_idle	out	1	ap_ctrl_hs	myproject	return value
ap_ready	out	1	ap_ctrl_hs	myproject	return value
input_1_ap_vld	in	1	ap_vld	input_1	pointer
input_1	in	160	ap_vld	input_1	pointer
layer5_out	out	16	ap_vld	layer5_out	pointer
layer5_out_ap_vld	out	1	ap_vld	layer5_out	pointer

KERAS 3_layer:

Report time : Sat Aug 29 02:12:20 PM EDT 2026.

Solution : solution1.

Simulation tool : xsim.

		Latency(Clock Cycles)			Interval(Clock Cycles)		
Total Execution Time							
+ RTL	+ Status						
(Clock Cycles)							
		min	avg	max	min	avg	max

VHDL	NA	NA	NA	NA	NA	NA	NA
NA							
Verilog	Pass	NA	NA	NA	NA	NA	NA
NA							
+-----+-----+-----+-----+-----+-----+-----+							
-----+							

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== Vitis HLS Report for 'myproject'

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* Date: Sat Aug 29 14:11:28 2026

* Version: 2025.2.1 (Build 6397637 on Mar 18 2026)

* Project: myproject_prj

* Solution: solution1 (Vivado IP Flow Target)

* Product family: virtexuplus

* Target device: xcvu13p-flga2577-2-e

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== Performance Estimates

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+ Timing:

* Summary:

+-----+-----+-----+-----+				
Clock Target Estimated Uncertainty				
+-----+-----+-----+-----+				
ap_clk 5.00 ns 3.490 ns 1.35 ns				
+-----+-----+-----+-----+				

+ Latency:

* Summary:

+-----+-----+-----+-----+-----+-----+							
Latency (cycles) Latency (absolute) Interval Pipeline							
min max min max min max Type							
+-----+-----+-----+-----+-----+-----+							
7 7 35.000 ns 35.000 ns 1 1 yes							
+-----+-----+-----+-----+-----+-----+							

+ Detail:

* Instance:

Latency (cycles)		Latency (absolute)		Interval		Pipeline			
min max		Instance						Module	
min max		min max		min max		Type			
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s_fu_49									
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s								1	1
5.000 ns		5.000 ns		1 1		yes			
call_ret2_relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s_fu_55									
relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s								0	0 0 ns
0 ns		1 1		yes					
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91									
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s								1	1
5.000 ns		5.000 ns		1 1		yes			
grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127									
sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s								2	2
10.000 ns		10.000 ns		1 1		yes			

* Loop:

N/A

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== Utilization Estimates

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* Summary:

Name		BRAM_18K		DSP		FF		LUT	
DSP		- -		- -		- -			
Expression		- -		0 10		-			
FIFO		- -		- -		- -			
Instance		1 305		1041 10551		-			
Memory		- -		- -		- -			
Multiplexer		- -		0 36		-			

Register		-	-	1212	-	-
+-----+-----+-----+-----+-----+						
Total		1	305	2253	10597	0
+-----+-----+-----+-----+-----+						
Available SLR		1344	3072	864000	432000	320
+-----+-----+-----+-----+-----+						
Utilization SLR (%)		~0	9	~0	2	0
+-----+-----+-----+-----+-----+						
Available		5376	12288	3456000	1728000	1280
+-----+-----+-----+-----+-----+						
Utilization (%)		~0	2	~0	~0	0
+-----+-----+-----+-----+-----+						

+ Detail:

* Instance:

+-----+-----+-----+-----+-----+						
-----+-----+-----+-----+-----+						
	Instance					Module
	BRAM_18K DSP FF LUT URAM					
+-----+-----+-----+-----+-----+						
-----+-----+-----+-----+-----+						
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s_fu_49						
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s 0 275 850						
8425 0						
grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91						
dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s 0 30 178						
797 0						
call_ret2_relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s_fu_55						
relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s 0 0 0 1216						
0						
grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127						
sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s 1 0 13						
113 0						
+-----+-----+-----+-----+-----+						
-----+-----+-----+-----+-----+						
Total						
	1 305 1041 10551 0					
+-----+-----+-----+-----+-----+						
-----+-----+-----+-----+-----+						

* DSP:

N/A

* Memory:

N/A

* FIFO:

N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
ap_block_pp0_stage0_11001	and	0	0	2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp10	and	0	0	2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp77	and	0	0	2	1	1
ap_block_pp0_stage0_11001_ignoreCallOp79	and	0	0	2	1	1
ap_enable_pp0	xor	0	0	2	1	2
Total		0	0	10	5	6

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
input_1_ap_vld_in_sig	9	2	1	2
input_1_ap_vld_preg	9	2	1	2
input_1_blk_n	9	2	1	2
input_1_in_sig	9	2	160	320
Total	36	8	163	326

* Register:

Name	FF	LUT	Bits	Const Bits
ap_CS_fsm	1	0	1	0
ap_block_pp0_stage0_subdone_grp0_done_reg		1	0	1
ap_enable_reg_pp0_iter1	1	0	1	0

ap_enable_reg_pp0_iter2		1	0	1	0
ap_enable_reg_pp0_iter3		1	0	1	0
ap_enable_reg_pp0_iter4		1	0	1	0
ap_enable_reg_pp0_iter5		1	0	1	0
ap_enable_reg_pp0_iter6		1	0	1	0
ap_enable_reg_pp0_iter7		1	0	1	0

grp_dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s_fu_91_ap_start_reg		1	0	1	0
--	--	---	---	---	---

grp_sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s_fu_127_ap_start_reg		1	0	1	0
---	--	---	---	---	---

input_1_ap_vld_preg		1	0	1	0
input_1_preg		160	0	160	0
layer2_out_10_reg_445		16	0	16	0
layer2_out_11_reg_450		16	0	16	0
layer2_out_12_reg_455		16	0	16	0
layer2_out_13_reg_460		16	0	16	0
layer2_out_14_reg_465		16	0	16	0
layer2_out_15_reg_470		16	0	16	0
layer2_out_16_reg_475		16	0	16	0
layer2_out_17_reg_480		16	0	16	0
layer2_out_18_reg_485		16	0	16	0
layer2_out_19_reg_490		16	0	16	0
layer2_out_1_reg_400		16	0	16	0
layer2_out_20_reg_495		16	0	16	0
layer2_out_21_reg_500		16	0	16	0
layer2_out_22_reg_505		16	0	16	0
layer2_out_23_reg_510		16	0	16	0
layer2_out_24_reg_515		16	0	16	0
layer2_out_25_reg_520		16	0	16	0
layer2_out_26_reg_525		16	0	16	0
layer2_out_27_reg_530		16	0	16	0
layer2_out_28_reg_535		16	0	16	0
layer2_out_29_reg_540		16	0	16	0
layer2_out_2_reg_405		16	0	16	0
layer2_out_30_reg_545		16	0	16	0
layer2_out_31_reg_550		16	0	16	0
layer2_out_3_reg_410		16	0	16	0
layer2_out_4_reg_415		16	0	16	0
layer2_out_5_reg_420		16	0	16	0
layer2_out_6_reg_425		16	0	16	0
layer2_out_7_reg_430		16	0	16	0
layer2_out_8_reg_435		16	0	16	0

layer2_out_9_reg_440	16 0 16	0
layer2_out_reg_395	16 0 16	0
layer3_out_10_reg_605	16 0 16	0
layer3_out_11_reg_610	16 0 16	0
layer3_out_12_reg_615	16 0 16	0
layer3_out_13_reg_620	16 0 16	0
layer3_out_14_reg_625	16 0 16	0
layer3_out_15_reg_630	16 0 16	0
layer3_out_16_reg_635	16 0 16	0
layer3_out_17_reg_640	16 0 16	0
layer3_out_18_reg_645	16 0 16	0
layer3_out_19_reg_650	16 0 16	0
layer3_out_1_reg_560	16 0 16	0
layer3_out_20_reg_655	16 0 16	0
layer3_out_21_reg_660	16 0 16	0
layer3_out_22_reg_665	16 0 16	0
layer3_out_23_reg_670	16 0 16	0
layer3_out_24_reg_675	16 0 16	0
layer3_out_25_reg_680	16 0 16	0
layer3_out_26_reg_685	16 0 16	0
layer3_out_27_reg_690	16 0 16	0
layer3_out_28_reg_695	16 0 16	0
layer3_out_29_reg_700	16 0 16	0
layer3_out_2_reg_565	16 0 16	0
layer3_out_30_reg_705	16 0 16	0
layer3_out_31_reg_710	16 0 16	0
layer3_out_3_reg_570	16 0 16	0
layer3_out_4_reg_575	16 0 16	0
layer3_out_5_reg_580	16 0 16	0
layer3_out_6_reg_585	16 0 16	0
layer3_out_7_reg_590	16 0 16	0
layer3_out_8_reg_595	16 0 16	0
layer3_out_9_reg_600	16 0 16	0
layer3_out_reg_555	16 0 16	0
layer4_out_reg_715	16 0 16	0

-----+-----+-----+-----+		
Total	1212 0 1212	0

-----+-----+-----+-----+

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== Interface

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* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
ap_clk	in	1	ap_ctrl_hs	myproject	return value
ap_rst	in	1	ap_ctrl_hs	myproject	return value
ap_start	in	1	ap_ctrl_hs	myproject	return value
ap_done	out	1	ap_ctrl_hs	myproject	return value
ap_idle	out	1	ap_ctrl_hs	myproject	return value
ap_ready	out	1	ap_ctrl_hs	myproject	return value
input_1_ap_vld	in	1	ap_vld	input_1	pointer
input_1	in	160	ap_vld	input_1	pointer
layer5_out	out	16	ap_vld	layer5_out	pointer
layer5_out_ap_vld	out	1	ap_vld	layer5_out	pointer

+

KERAS 3 layer batch norm:

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== Synthesis Summary Report of 'myproject'

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+ General Information:

* Date: Sat Aug 29 14:22:14 2026

* Version: 2025.2.1 (Build 6397637 on Mar 18 2026)

* Project: myproject_prj

* Solution: solution1 (Vivado IP Flow Target)

* Product family: virtexuplus

* Target device: xcvu13p-flga2577-2-e

* Performance & Resource Estimates:

Latency	Count	Pipelined	Modules	(cycles)	(ns)	Slack	BRAM	DSP	FF	LUT	URAM
			& Loops								

+

+

+ myproject		-	-	-	1	-	yes
7 35.000 0.16 1 (~0%) 305 (2%) 2253 (~0%) 10597 (~0%) -							
+ dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s		-					-
- 1 - yes 1 5.000 0.16 - 275 (2%) 850 (~0%) 8425 (~0%) -							
+ relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s		-					-
- 1 - yes 0 0.000 2.57 - - - 1216 (~0%) -							
+ dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s		-					-
- 1 - yes 1 5.000 0.25 - 30 (~0%) 178 (~0%) 797 (~0%) -							
+ sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s		-					-
- 1 - yes 2 10.000 1.12 1 (~0%) - 13 (~0%) 113 (~0%) -							
+-----+							
+-----+							

Name Prefix: '+' for module, 'o' for loop, '*' for dataflow

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== HW Interfaces

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* Other Ports

Port	Mode	Direction	Bitwidth
input_1	ap_vld	in	160
layer5_out	ap_vld	out	16

* TOP LEVEL CONTROL

Interface	Type	Ports
ap_clk	clock	ap_clk
ap_rst	reset	ap_rst
ap_ctrl	ap_ctrl_hs	ap_done ap_idle ap_ready ap_start

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== SW I/O Information

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* Top Function Arguments

Argument	Direction	Datatype
input_1	in	ap_fixed<16, 6, (ap_q_mode)5, (ap_o_mode)3, 0>*

layer5_out	out	ap_fixed<16, 6, (ap_q_mode)5, (ap_o_mode)3, 0>*
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* SW-to-HW Mapping

Argument	HW Interface	HW Type
input_1	input_1_ap_vld	port
input_1	input_1	port
layer5_out	layer5_out	port
layer5_out	layer5_out_ap_vld	port

```
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```

== Bind Op Report

```
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```

Name	DSP	Pragma	Variable	Op	Impl
Latency					
+ myproject	305				
+ dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config2_s	275				
sub_ln73_fu_283059_p2			sub_ln73	sub	fabric
mul_16s_9s_25_1_1_U95	1		mul_ln73	mul	auto
add_ln73_fu_283104_p2			add_ln73	add	fabric
mul_16s_7s_23_1_1_U171	1		mul_ln73_12	mul	
sub_ln73_2_fu_283160_p2			sub_ln73_2	sub	
mul_16s_11ns_26_1_1_U92	1		mul_ln42	mul	
mul_16s_7ns_23_1_1_U80	1		mul_ln73_13	mul	
add_ln73_1_fu_283215_p2			add_ln73_1	add	
mul_16s_10ns_26_1_1_U1	1		mul_ln73_14	mul	

mul_16s_8s_24_1_1_U157 auto 0	1 mul_ln73_15 mul
mul_16s_8s_24_1_1_U13 auto 0	1 mul_ln73_16 mul
mul_16s_8ns_24_1_1_U159 auto 0	1 mul_ln73_17 mul
mul_16s_9ns_25_1_1_U272 auto 0	1 mul_ln73_18 mul
mul_16s_6ns_22_1_1_U142 auto 0	1 mul_ln73_19 mul
mul_16s_10ns_26_1_1_U232 auto 0	1 mul_ln73_20 mul
mul_16s_7s_23_1_1_U233 auto 0	1 mul_ln73_21 mul
sub_ln73_5_fu_283380_p2 fabric 0	sub_ln73_5 sub
mul_16s_10s_26_1_1_U172 auto 0	1 mul_ln73_22 mul
mul_16s_9ns_25_1_1_U32 auto 0	1 mul_ln73_23 mul
mul_16s_10ns_26_1_1_U181 auto 0	1 mul_ln73_24 mul
sub_ln73_6_fu_283449_p2 fabric 0	sub_ln73_6 sub
mul_16s_11s_26_1_1_U42 auto 0	1 mul_ln42_18 mul
mul_16s_9ns_25_1_1_U52 auto 0	1 mul_ln73_25 mul
sub_ln73_7_fu_283497_p2 fabric 0	sub_ln73_7 sub
mul_16s_8ns_24_1_1_U196 auto 0	1 mul_ln73_26 mul
mul_16s_10ns_26_1_1_U239 auto 0	1 mul_ln73_27 mul
mul_16s_11s_26_1_1_U56 auto 0	1 mul_ln42_19 mul
mul_16s_10ns_26_1_1_U9 auto 0	1 mul_ln73_28 mul
mul_16s_9ns_25_1_1_U158 auto 0	1 mul_ln73_29 mul
mul_16s_11ns_26_1_1_U12 auto 0	1 mul_ln42_20 mul
mul_16s_9s_25_1_1_U112 auto 0	1 mul_ln73_30 mul

mul_16s_10ns_26_1_1_U227 auto 0	1	mul_ln73_31 mul
mul_16s_11ns_26_1_1_U120 auto 0	1	mul_ln42_21 mul
mul_16s_8s_24_1_1_U70 auto 0	1	mul_ln73_32 mul
mul_16s_8ns_24_1_1_U69 auto 0	1	mul_ln73_33 mul
mul_16s_9ns_25_1_1_U15 auto 0	1	mul_ln73_34 mul
mul_16s_7ns_23_1_1_U167 auto 0	1	mul_ln73_35 mul
mul_16s_5s_21_1_1_U253 auto 0	1	mul_ln73_36 mul
mul_16s_10ns_26_1_1_U26 auto 0	1	mul_ln73_37 mul
mul_16s_9ns_25_1_1_U200 auto 0	1	mul_ln73_38 mul
sub_ln73_8_fu_283781_p2 fabric 0		sub_ln73_8 sub
mul_16s_11ns_26_1_1_U271 auto 0	1	mul_ln42_22 mul
mul_16s_11ns_26_1_1_U258 auto 0	1	mul_ln42_23 mul
mul_16s_9ns_25_1_1_U2 auto 0	1	mul_ln73_39 mul
mul_16s_9ns_25_1_1_U116 auto 0	1	mul_ln73_40 mul
mul_16s_11ns_26_1_1_U174 auto 0	1	mul_ln42_24 mul
mul_16s_7s_23_1_1_U35 auto 0	1	mul_ln73_41 mul
sub_ln73_9_fu_283873_p2 fabric 0		sub_ln73_9 sub
mul_16s_10s_26_1_1_U130 auto 0	1	mul_ln73_42 mul
mul_16s_9s_25_1_1_U133 auto 0	1	mul_ln73_43 mul
mul_16s_9ns_25_1_1_U175 auto 0	1	mul_ln73_44 mul
mul_16s_9s_25_1_1_U111 auto 0	1	mul_ln73_45 mul
mul_16s_9ns_25_1_1_U115 auto 0	1	mul_ln73_46 mul

mul_16s_11ns_26_1_1_U27 auto 0	1	mul_ln42_25 mul
mul_16s_7s_23_1_1_U55 auto 0	1	mul_ln73_47 mul
mul_16s_10s_26_1_1_U94 auto 0	1	mul_ln73_48 mul
mul_16s_9ns_25_1_1_U228 auto 0	1	mul_ln73_49 mul
mul_16s_11ns_26_1_1_U273 auto 0	1	mul_ln42_26 mul
add_ln73_2_fu_284076_p2 fabric 0		add_ln73_2 add
mul_16s_11s_26_1_1_U186 auto 0	1	mul_ln42_27 mul
mul_16s_10s_26_1_1_U265 auto 0	1	mul_ln73_50 mul
mul_16s_9ns_25_1_1_U139 auto 0	1	mul_ln73_51 mul
mul_16s_12s_26_1_1_U47 auto 0	1	mul_ln42_28 mul
mul_16s_11s_26_1_1_U147 auto 0	1	mul_ln42_29 mul
mul_16s_12s_26_1_1_U149 auto 0	1	mul_ln42_30 mul
mul_16s_9ns_25_1_1_U83 auto 0	1	mul_ln73_52 mul
mul_16s_10s_26_1_1_U229 auto 0	1	mul_ln73_53 mul
mul_16s_11ns_26_1_1_U97 auto 0	1	mul_ln42_31 mul
mul_16s_10s_26_1_1_U145 auto 0	1	mul_ln73_54 mul
mul_16s_10s_26_1_1_U192 auto 0	1	mul_ln73_55 mul
mul_16s_6ns_22_1_1_U103 auto 0	1	mul_ln73_56 mul
mul_16s_12s_26_1_1_U98 auto 0	1	mul_ln42_32 mul
mul_16s_10s_26_1_1_U238 auto 0	1	mul_ln73_57 mul
mul_16s_11s_26_1_1_U201 auto 0	1	mul_ln42_33 mul
mul_16s_9s_25_1_1_U207 auto 0	1	mul_ln73_58 mul

mul_16s_10s_26_1_1_U155 auto 0	1	mul_ln73_59 mul
mul_16s_10s_26_1_1_U107 auto 0	1	mul_ln73_60 mul
mul_16s_10ns_26_1_1_U199 auto 0	1	mul_ln73_61 mul
mul_16s_11ns_26_1_1_U247 auto 0	1	mul_ln42_34 mul
mul_16s_9ns_25_1_1_U64 auto 0	1	mul_ln73_62 mul
mul_16s_11s_26_1_1_U79 auto 0	1	mul_ln42_35 mul
mul_16s_10ns_26_1_1_U197 auto 0	1	mul_ln73_63 mul
mul_16s_11ns_26_1_1_U3 auto 0	1	mul_ln42_36 mul
mul_16s_10ns_26_1_1_U90 auto 0	1	mul_ln73_64 mul
mul_16s_9s_25_1_1_U24 auto 0	1	mul_ln73_65 mul
mul_16s_11ns_26_1_1_U250 auto 0	1	mul_ln42_37 mul
mul_16s_11s_26_1_1_U19 auto 0	1	mul_ln42_38 mul
mul_16s_11ns_26_1_1_U212 auto 0	1	mul_ln42_39 mul
mul_16s_10ns_26_1_1_U214 auto 0	1	mul_ln73_66 mul
mul_16s_10ns_26_1_1_U254 auto 0	1	mul_ln73_67 mul
mul_16s_9s_25_1_1_U195 auto 0	1	mul_ln73_68 mul
sub_ln73_12_fu_284505_p2 fabric 0		sub_ln73_12 sub
mul_16s_9s_25_1_1_U129 auto 0	1	mul_ln73_69 mul
sub_ln73_13_fu_284561_p2 fabric 0		sub_ln73_13 sub
mul_16s_11s_26_1_1_U63 auto 0	1	mul_ln42_40 mul
mul_16s_11s_26_1_1_U148 auto 0	1	mul_ln42_41 mul
mul_16s_8ns_24_1_1_U93 auto 0	1	mul_ln73_70 mul

mul_16s_9s_25_1_1_U82 auto 0	1 mul_ln73_71 mul
mul_16s_9s_25_1_1_U262 auto 0	1 mul_ln73_72 mul
mul_16s_11ns_26_1_1_U223 auto 0	1 mul_ln42_42 mul
mul_16s_10ns_26_1_1_U40 auto 0	1 mul_ln73_73 mul
mul_16s_8s_24_1_1_U237 auto 0	1 mul_ln73_74 mul
mul_16s_9ns_25_1_1_U104 auto 0	1 mul_ln73_75 mul
mul_16s_10s_26_1_1_U160 auto 0	1 mul_ln73_76 mul
mul_16s_6ns_22_1_1_U220 auto 0	1 mul_ln73_77 mul
mul_16s_11s_26_1_1_U14 auto 0	1 mul_ln42_43 mul
mul_16s_11s_26_1_1_U184 auto 0	1 mul_ln42_44 mul
mul_16s_9s_25_1_1_U44 auto 0	1 mul_ln73_78 mul
mul_16s_10ns_26_1_1_U96 auto 0	1 mul_ln73_79 mul
mul_16s_10ns_26_1_1_U274 auto 0	1 mul_ln73_80 mul
mul_16s_10ns_26_1_1_U275 auto 0	1 mul_ln73_81 mul
mul_16s_11s_26_1_1_U146 auto 0	1 mul_ln42_45 mul
mul_16s_11s_26_1_1_U263 auto 0	1 mul_ln42_46 mul
mul_16s_10s_26_1_1_U268 auto 0	1 mul_ln73_82 mul
mul_16s_10ns_26_1_1_U45 auto 0	1 mul_ln73_83 mul
mul_16s_11s_26_1_1_U59 auto 0	1 mul_ln42_47 mul
add_ln73_3_fu_284875_p2 fabric 0	add_ln73_3 add
mul_16s_10s_26_1_1_U140 auto 0	1 mul_ln73_84 mul
mul_16s_10ns_26_1_1_U144 auto 0	1 mul_ln73_85 mul

mul_16s_6s_22_1_1_U50 auto 0	1 mul_ln73_86 mul
mul_16s_11ns_26_1_1_U198 auto 0	1 mul_ln42_48 mul
sub_ln73_16_fu_284980_p2 fabric 0	sub_ln73_16 sub
mul_16s_11s_26_1_1_U109 auto 0	1 mul_ln42_49 mul
mul_16s_9s_25_1_1_U235 auto 0	1 mul_ln73_87 mul
mul_16s_7ns_23_1_1_U241 auto 0	1 mul_ln73_88 mul
mul_16s_10ns_26_1_1_U248 auto 0	1 mul_ln73_89 mul
mul_16s_9s_25_1_1_U43 auto 0	1 mul_ln73_90 mul
mul_16s_11ns_26_1_1_U46 auto 0	1 mul_ln42_50 mul
mul_16s_9ns_25_1_1_U110 auto 0	1 mul_ln73_91 mul
mul_16s_10s_26_1_1_U208 auto 0	1 mul_ln73_92 mul
mul_16s_9ns_25_1_1_U162 auto 0	1 mul_ln73_93 mul
mul_16s_10ns_26_1_1_U194 auto 0	1 mul_ln73_94 mul
mul_16s_11ns_26_1_1_U118 auto 0	1 mul_ln42_51 mul
mul_16s_10s_26_1_1_U264 auto 0	1 mul_ln73_95 mul
sub_ln73_17_fu_285203_p2 fabric 0	sub_ln73_17 sub
mul_16s_10s_26_1_1_U10 auto 0	1 mul_ln73_96 mul
mul_16s_10ns_26_1_1_U211 auto 0	1 mul_ln73_97 mul
mul_16s_10ns_26_1_1_U215 auto 0	1 mul_ln73_98 mul
mul_16s_9ns_25_1_1_U41 auto 0	1 mul_ln73_99 mul
mul_16s_10s_26_1_1_U210 auto 0	1 mul_ln73_100 mul
mul_16s_9ns_25_1_1_U251 auto 0	1 mul_ln73_101 mul

mul_16s_11s_26_1_1_U168 auto 0	1	mul_ln42_52 mul
mul_16s_9ns_25_1_1_U170 auto 0	1	mul_ln73_102 mul
mul_16s_9ns_25_1_1_U81 auto 0	1	mul_ln73_103 mul
mul_16s_9ns_25_1_1_U121 auto 0	1	mul_ln73_104 mul
mul_16s_10s_26_1_1_U185 auto 0	1	mul_ln73_105 mul
mul_16s_9s_25_1_1_U257 auto 0	1	mul_ln73_106 mul
mul_16s_8ns_24_1_1_U134 auto 0	1	mul_ln73_107 mul
mul_16s_10s_26_1_1_U66 auto 0	1	mul_ln73_108 mul
mul_16s_9s_25_1_1_U259 auto 0	1	mul_ln73_109 mul
mul_16s_8s_24_1_1_U222 auto 0	1	mul_ln73_110 mul
mul_16s_10s_26_1_1_U39 auto 0	1	mul_ln73_111 mul
mul_16s_10ns_26_1_1_U225 auto 0	1	mul_ln73_112 mul
sub_ln73_20_fu_285487_p2 fabric 0		sub_ln73_20 sub
mul_16s_9s_25_1_1_U226 auto 0	1	mul_ln73_113 mul
sub_ln73_21_fu_279583_p2 fabric 0		sub_ln73_21 sub
mul_16s_10s_26_1_1_U37 auto 0	1	mul_ln73_114 mul
mul_16s_10s_26_1_1_U71 auto 0	1	mul_ln73_115 mul
mul_16s_11ns_26_1_1_U209 auto 0	1	mul_ln42_53 mul
mul_16s_11s_26_1_1_U217 auto 0	1	mul_ln42_54 mul
add_ln73_4_fu_279645_p2 fabric 0		add_ln73_4 add
mul_16s_9s_25_1_1_U151 auto 0	1	mul_ln73_116 mul
mul_16s_9s_25_1_1_U36 auto 0	1	mul_ln73_117 mul

mul_16s_9ns_25_1_1_U224 auto 0	1	mul_ln73_118 mul
mul_16s_9ns_25_1_1_U213 auto 0	1	mul_ln73_119 mul
add_ln73_5_fu_279737_p2 fabric 0		add_ln73_5 add
mul_16s_10ns_26_1_1_U84 auto 0	1	mul_ln73_120 mul
mul_16s_7s_23_1_1_U4 auto 0	1	mul_ln73_121 mul
mul_16s_9s_25_1_1_U270 auto 0	1	mul_ln73_122 mul
mul_16s_10s_26_1_1_U191 auto 0	1	mul_ln73_123 mul
mul_16s_10s_26_1_1_U61 auto 0	1	mul_ln73_124 mul
mul_16s_11ns_26_1_1_U119 auto 0	1	mul_ln42_55 mul
mul_16s_11ns_26_1_1_U163 auto 0	1	mul_ln42_56 mul
mul_16s_10s_26_1_1_U216 auto 0	1	mul_ln73_125 mul
mul_16s_11ns_26_1_1_U99 auto 0	1	mul_ln42_57 mul
mul_16s_8ns_24_1_1_U266 auto 0	1	mul_ln73_126 mul
mul_16s_10ns_26_1_1_U68 auto 0	1	mul_ln73_127 mul
mul_16s_10ns_26_1_1_U122 auto 0	1	mul_ln73_128 mul
mul_16s_9s_25_1_1_U67 auto 0	1	mul_ln73_129 mul
mul_16s_10ns_26_1_1_U255 auto 0	1	mul_ln73_130 mul
sub_ln73_24_fu_279955_p2 fabric 0		sub_ln73_24 sub
mul_16s_10s_26_1_1_U173 auto 0	1	mul_ln73_131 mul
mul_16s_10s_26_1_1_U114 auto 0	1	mul_ln73_132 mul
sub_ln73_25_fu_279997_p2 fabric 0		sub_ln73_25 sub
mul_16s_7s_23_1_1_U132 auto 0	1	mul_ln73_133 mul

mul_16s_11ns_26_1_1_U230 auto 0	1	mul_ln42_58 mul
mul_16s_10s_26_1_1_U29 auto 0	1	mul_ln73_134 mul
mul_16s_9s_25_1_1_U74 auto 0	1	mul_ln73_135 mul
mul_16s_10ns_26_1_1_U123 auto 0	1	mul_ln73_136 mul
mul_16s_7s_23_1_1_U245 auto 0	1	mul_ln73_137 mul
mul_16s_10ns_26_1_1_U261 auto 0	1	mul_ln73_138 mul
mul_16s_9ns_25_1_1_U62 auto 0	1	mul_ln73_139 mul
mul_16s_10ns_26_1_1_U51 auto 0	1	mul_ln73_140 mul
mul_16s_10ns_26_1_1_U204 auto 0	1	mul_ln73_141 mul
mul_16s_11ns_26_1_1_U5 auto 0	1	mul_ln42_59 mul
mul_16s_11ns_26_1_1_U187 auto 0	1	mul_ln42_60 mul
mul_16s_9ns_25_1_1_U91 auto 0	1	mul_ln73_142 mul
mul_16s_8s_24_1_1_U11 auto 0	1	mul_ln73_143 mul
mul_16s_10ns_26_1_1_U202 auto 0	1	mul_ln73_144 mul
mul_16s_10s_26_1_1_U126 auto 0	1	mul_ln73_145 mul
mul_16s_11s_26_1_1_U127 auto 0	1	mul_ln42_61 mul
mul_16s_11s_26_1_1_U106 auto 0	1	mul_ln42_62 mul
mul_16s_11ns_26_1_1_U117 auto 0	1	mul_ln42_63 mul
mul_16s_11ns_26_1_1_U105 auto 0	1	mul_ln42_64 mul
mul_16s_10s_26_1_1_U152 auto 0	1	mul_ln73_146 mul
mul_16s_9ns_25_1_1_U153 auto 0	1	mul_ln73_147 mul
mul_16s_9ns_25_1_1_U246 auto 0	1	mul_ln73_148 mul

mul_16s_9ns_25_1_1_U89 auto 0	1	mul_ln73_149 mul
mul_16s_11ns_26_1_1_U182 auto 0	1	mul_ln42_65 mul
mul_16s_9s_25_1_1_U53 auto 0	1	mul_ln73_150 mul
add_ln73_6_fu_280442_p2 fabric 0		add_ln73_6 add
mul_16s_11ns_26_1_1_U178 auto 0	1	mul_ln42_66 mul
mul_16s_10s_26_1_1_U179 auto 0	1	mul_ln73_151 mul
mul_16s_9s_25_1_1_U252 auto 0	1	mul_ln73_152 mul
sub_ln42_fu_280504_p2 0		sub_ln42 sub fabric
add_ln42_fu_280602_p2 0		add_ln42 add fabric
mul_16s_10ns_26_1_1_U28 auto 0	1	mul_ln73_153 mul
mul_16s_11ns_26_1_1_U161 auto 0	1	mul_ln42_67 mul
mul_16s_9s_25_1_1_U100 auto 0	1	mul_ln73_154 mul
mul_16s_9ns_25_1_1_U177 auto 0	1	mul_ln73_155 mul
sub_ln73_28_fu_280666_p2 fabric 0		sub_ln73_28 sub
mul_16s_9s_25_1_1_U102 auto 0	1	mul_ln73_156 mul
mul_16s_9s_25_1_1_U249 auto 0	1	mul_ln73_157 mul
sub_ln73_31_fu_280768_p2 fabric 0		sub_ln73_31 sub
mul_16s_11ns_26_1_1_U49 auto 0	1	mul_ln42_68 mul
mul_16s_9ns_25_1_1_U65 auto 0	1	mul_ln73_158 mul
sub_ln73_32_fu_280820_p2 fabric 0		sub_ln73_32 sub
mul_16s_8ns_24_1_1_U21 auto 0	1	mul_ln73_159 mul
sub_ln73_33_fu_280866_p2 fabric 0		sub_ln73_33 sub

mul_16s_9ns_25_1_1_U231 auto 0	1	mul_ln73_160 mul
mul_16s_10s_26_1_1_U135 auto 0	1	mul_ln73_161 mul
add_ln73_7_fu_280926_p2 fabric 0		add_ln73_7 add
sub_ln73_34_fu_280950_p2 fabric 0		sub_ln73_34 sub
mul_16s_11ns_26_1_1_U156 auto 0	1	mul_ln42_69 mul
mul_16s_10s_26_1_1_U143 auto 0	1	mul_ln73_162 mul
mul_16s_10ns_26_1_1_U58 auto 0	1	mul_ln73_163 mul
mul_16s_9ns_25_1_1_U256 auto 0	1	mul_ln73_164 mul
mul_16s_10s_26_1_1_U242 auto 0	1	mul_ln73_165 mul
mul_16s_10ns_26_1_1_U85 auto 0	1	mul_ln73_166 mul
mul_16s_10s_26_1_1_U221 auto 0	1	mul_ln73_167 mul
mul_16s_9ns_25_1_1_U18 auto 0	1	mul_ln73_168 mul
mul_16s_10s_26_1_1_U75 auto 0	1	mul_ln73_169 mul
mul_16s_11ns_26_1_1_U7 auto 0	1	mul_ln42_70 mul
mul_16s_10s_26_1_1_U234 auto 0	1	mul_ln73_170 mul
mul_16s_10s_26_1_1_U137 auto 0	1	mul_ln73_171 mul
mul_16s_10s_26_1_1_U54 auto 0	1	mul_ln73_172 mul
add_ln42_1_fu_281194_p2 fabric 0		add_ln42_1 add
mul_16s_10ns_26_1_1_U73 auto 0	1	mul_ln73_173 mul
mul_16s_9ns_25_1_1_U205 auto 0	1	mul_ln73_174 mul
mul_16s_9ns_25_1_1_U206 auto 0	1	mul_ln73_175 mul
mul_16s_8s_24_1_1_U25 auto 0	1	mul_ln73_176 mul

| sub_ln73_35_fu_281278_p2
 fabric | 0 |
 | mul_16s_11ns_26_1_1_U87
 auto | 0 |
 | mul_16s_11ns_26_1_1_U76
 auto | 0 |
 | mul_16s_6s_22_1_1_U60
 auto | 0 |
 | mul_16s_10ns_26_1_1_U165
 auto | 0 |
 | mul_16s_11ns_26_1_1_U131
 auto | 0 |
 | mul_16s_12s_26_1_1_U193
 auto | 0 |
 | mul_16s_12ns_26_1_1_U183
 auto | 0 |
 | mul_16s_12s_26_1_1_U188
 auto | 0 |
 | mul_16s_11s_26_1_1_U31
 auto | 0 |
 | mul_16s_9s_25_1_1_U154
 auto | 0 |
 | mul_16s_7s_23_1_1_U23
 auto | 0 |
 | mul_16s_11ns_26_1_1_U141
 auto | 0 |
 | mul_16s_8ns_24_1_1_U57
 auto | 0 |
 | mul_16s_10s_26_1_1_U6
 auto | 0 |
 | mul_16s_10ns_26_1_1_U219
 auto | 0 |
 | mul_16s_11ns_26_1_1_U8
 auto | 0 |
 | sub_ln73_36_fu_281498_p2
 fabric | 0 |
 | sub_ln73_37_fu_281518_p2
 fabric | 0 |
 | mul_16s_10ns_26_1_1_U30
 auto | 0 |
 | mul_16s_10s_26_1_1_U16
 auto | 0 |
 | mul_16s_11s_26_1_1_U176
 auto | 0 |

		sub_ln73_35	sub
1		mul_ln42_71	mul
1		mul_ln42_72	mul
1		mul_ln73_177	mul
1		mul_ln73_178	mul
1		mul_ln42_73	mul
1		mul_ln42_74	mul
1		mul_ln42_75	mul
1		mul_ln42_76	mul
1		mul_ln42_77	mul
1		mul_ln73_179	mul
1		mul_ln73_180	mul
1		mul_ln42_78	mul
1		mul_ln73_181	mul
1		mul_ln73_182	mul
1		mul_ln73_183	mul
1		mul_ln42_79	mul
		sub_ln73_36	sub
		sub_ln73_37	sub
1		mul_ln73_184	mul
1		mul_ln73_185	mul
1		mul_ln42_80	mul

mul_16s_11s_26_1_1_U34 auto 0	1 mul_ln42_81 mul
mul_16s_10ns_26_1_1_U244 auto 0	1 mul_ln73_186 mul
mul_16s_11ns_26_1_1_U180 auto 0	1 mul_ln42_82 mul
mul_16s_10ns_26_1_1_U125 auto 0	1 mul_ln73_187 mul
mul_16s_12s_26_1_1_U113 auto 0	1 mul_ln42_83 mul
mul_16s_10s_26_1_1_U203 auto 0	1 mul_ln73_188 mul
mul_16s_11s_26_1_1_U269 auto 0	1 mul_ln42_84 mul
mul_16s_7s_23_1_1_U101 auto 0	1 mul_ln73_189 mul
sub_ln42_1_fu_281741_p2 fabric 0	sub_ln42_1 sub
mul_16s_10ns_26_1_1_U88 auto 0	1 mul_ln73_190 mul
mul_16s_9ns_25_1_1_U260 auto 0	1 mul_ln73_191 mul
mul_16s_10s_26_1_1_U77 auto 0	1 mul_ln73_192 mul
mul_16s_11s_26_1_1_U20 auto 0	1 mul_ln42_85 mul
mul_16s_11ns_26_1_1_U72 auto 0	1 mul_ln42_86 mul
mul_16s_11s_26_1_1_U236 auto 0	1 mul_ln42_87 mul
mul_16s_10s_26_1_1_U108 auto 0	1 mul_ln73_193 mul
mul_16s_12s_26_1_1_U169 auto 0	1 mul_ln42_88 mul
mul_16s_9s_25_1_1_U218 auto 0	1 mul_ln73_194 mul
mul_16s_9ns_25_1_1_U48 auto 0	1 mul_ln73_195 mul
mul_16s_10s_26_1_1_U243 auto 0	1 mul_ln73_196 mul
mul_16s_10s_26_1_1_U86 auto 0	1 mul_ln73_197 mul
mul_16s_11ns_26_1_1_U128 auto 0	1 mul_ln42_89 mul

mul_16s_11ns_26_1_1_U136 auto 0	1	mul_ln42_90 mul
mul_16s_8s_24_1_1_U150 auto 0	1	mul_ln73_198 mul
mul_16s_10s_26_1_1_U138 auto 0	1	mul_ln73_199 mul
mul_16s_10ns_26_1_1_U78 auto 0	1	mul_ln73_200 mul
mul_16s_11ns_26_1_1_U17 auto 0	1	mul_ln42_91 mul
mul_16s_10ns_26_1_1_U267 auto 0	1	mul_ln73_201 mul
mul_16s_11ns_26_1_1_U164 auto 0	1	mul_ln42_92 mul
mul_16s_10s_26_1_1_U33 auto 0	1	mul_ln73_202 mul
mul_16s_11ns_26_1_1_U166 auto 0	1	mul_ln42_93 mul
mul_16s_7ns_23_1_1_U240 auto 0	1	mul_ln73_203 mul
mul_16s_8ns_24_1_1_U38 auto 0	1	mul_ln73_204 mul
mul_16s_6s_22_1_1_U124 auto 0	1	mul_ln73_205 mul
mul_16s_9ns_25_1_1_U189 auto 0	1	mul_ln73_206 mul
mul_16s_9s_25_1_1_U190 auto 0	1	mul_ln73_207 mul
mul_16s_7ns_23_1_1_U22 auto 0	1	mul_ln73_208 mul
add_ln58_fu_285553_p2 0		add_ln58 add fabric
add_ln58_35_fu_282077_p2 fabric 0		add_ln58_35 add
add_ln58_41_fu_285582_p2 fabric 0		add_ln58_41 add
add_ln58_45_fu_282107_p2 fabric 0		add_ln58_45 add
add_ln58_51_fu_285611_p2 fabric 0		add_ln58_51 add
add_ln58_55_fu_282137_p2 fabric 0		add_ln58_55 add
add_ln58_57_fu_282143_p2 fabric 0		add_ln58_57 add

add_ln58_58_fu_282153_p2 fabric 0	add_ln58_58 add
add_ln58_61_fu_285655_p2 fabric 0	add_ln58_61 add
add_ln58_65_fu_282159_p2 fabric 0	add_ln58_65 add
add_ln58_71_fu_285688_p2 fabric 0	add_ln58_71 add
add_ln58_75_fu_282189_p2 fabric 0	add_ln58_75 add
add_ln58_81_fu_285717_p2 fabric 0	add_ln58_81 add
add_ln58_85_fu_282219_p2 fabric 0	add_ln58_85 add
add_ln58_86_fu_282229_p2 fabric 0	add_ln58_86 add
add_ln58_90_fu_285746_p2 fabric 0	add_ln58_90 add
add_ln58_94_fu_282247_p2 fabric 0	add_ln58_94 add
add_ln58_100_fu_285779_p2 fabric 0	add_ln58_100 add
add_ln58_104_fu_282277_p2 fabric 0	add_ln58_104 add
add_ln58_110_fu_285808_p2 fabric 0	add_ln58_110 add
add_ln58_114_fu_282307_p2 fabric 0	add_ln58_114 add
add_ln58_120_fu_285837_p2 fabric 0	add_ln58_120 add
add_ln58_124_fu_282337_p2 fabric 0	add_ln58_124 add
add_ln58_130_fu_285870_p2 fabric 0	add_ln58_130 add
add_ln58_131_fu_285876_p2 fabric 0	add_ln58_131 add
add_ln58_132_fu_285886_p2 fabric 0	add_ln58_132 add
add_ln58_134_fu_282367_p2 fabric 0	add_ln58_134 add
add_ln58_140_fu_285903_p2 fabric 0	add_ln58_140 add
add_ln58_144_fu_282397_p2 fabric 0	add_ln58_144 add

add_ln58_150_fu_285932_p2 fabric 0	add_ln58_150 add
add_ln58_154_fu_282427_p2 fabric 0	add_ln58_154 add
add_ln58_160_fu_285961_p2 fabric 0	add_ln58_160 add
add_ln58_164_fu_282457_p2 fabric 0	add_ln58_164 add
add_ln58_170_fu_285990_p2 fabric 0	add_ln58_170 add
add_ln58_174_fu_282491_p2 fabric 0	add_ln58_174 add
add_ln58_180_fu_286019_p2 fabric 0	add_ln58_180 add
add_ln58_184_fu_282521_p2 fabric 0	add_ln58_184 add
add_ln58_190_fu_286048_p2 fabric 0	add_ln58_190 add
add_ln58_194_fu_282551_p2 fabric 0	add_ln58_194 add
add_ln58_200_fu_286081_p2 fabric 0	add_ln58_200 add
add_ln58_204_fu_282581_p2 fabric 0	add_ln58_204 add
add_ln58_210_fu_286110_p2 fabric 0	add_ln58_210 add
add_ln58_214_fu_282611_p2 fabric 0	add_ln58_214 add
add_ln58_216_fu_282617_p2 fabric 0	add_ln58_216 add
add_ln58_217_fu_282627_p2 fabric 0	add_ln58_217 add
add_ln58_220_fu_286150_p2 fabric 0	add_ln58_220 add
add_ln58_224_fu_282633_p2 fabric 0	add_ln58_224 add
add_ln58_230_fu_286179_p2 fabric 0	add_ln58_230 add
add_ln58_234_fu_282663_p2 fabric 0	add_ln58_234 add
add_ln58_236_fu_282675_p2 fabric 0	add_ln58_236 add
add_ln58_237_fu_282685_p2 fabric 0	add_ln58_237 add

add_ln58_240_fu_286208_p2 fabric 0	add_ln58_240 add
add_ln58_244_fu_282697_p2 fabric 0	add_ln58_244 add
add_ln58_250_fu_286237_p2 fabric 0	add_ln58_250 add
add_ln58_254_fu_282727_p2 fabric 0	add_ln58_254 add
add_ln58_260_fu_286266_p2 fabric 0	add_ln58_260 add
add_ln58_264_fu_282757_p2 fabric 0	add_ln58_264 add
add_ln58_270_fu_286295_p2 fabric 0	add_ln58_270 add
add_ln58_274_fu_282787_p2 fabric 0	add_ln58_274 add
add_ln58_280_fu_286324_p2 fabric 0	add_ln58_280 add
add_ln58_284_fu_282817_p2 fabric 0	add_ln58_284 add
add_ln58_290_fu_286353_p2 fabric 0	add_ln58_290 add
add_ln58_294_fu_282847_p2 fabric 0	add_ln58_294 add
add_ln58_296_fu_282853_p2 fabric 0	add_ln58_296 add
add_ln58_297_fu_282863_p2 fabric 0	add_ln58_297 add
add_ln58_300_fu_286393_p2 fabric 0	add_ln58_300 add
add_ln58_304_fu_282869_p2 fabric 0	add_ln58_304 add
add_ln58_306_fu_282875_p2 fabric 0	add_ln58_306 add
add_ln58_307_fu_282885_p2 fabric 0	add_ln58_307 add
add_ln58_310_fu_286433_p2 fabric 0	add_ln58_310 add
add_ln58_314_fu_282891_p2 fabric 0	add_ln58_314 add
add_ln58_316_fu_282897_p2 fabric 0	add_ln58_316 add
add_ln58_317_fu_282907_p2 fabric 0	add_ln58_317 add

add_ln58_320_fu_286473_p2 fabric 0	add_ln58_320 add
add_ln58_324_fu_282913_p2 fabric 0	add_ln58_324 add
add_ln58_330_fu_286502_p2 fabric 0	add_ln58_330 add
add_ln58_334_fu_282943_p2 fabric 0	add_ln58_334 add
add_ln58_340_fu_286531_p2 fabric 0	add_ln58_340 add
add_ln58_344_fu_282973_p2 fabric 0	add_ln58_344 add
add_ln58_346_fu_282979_p2 fabric 0	add_ln58_346 add
add_ln58_347_fu_282989_p2 fabric 0	add_ln58_347 add
+ relu_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_relu_config3_s 0	
icmp_ln45_fu_400_p2 0	icmp_ln45 setgt auto
select_ln45_fu_406_p3 auto_sel 0	select_ln45 select
icmp_ln45_1_fu_418_p2 0	icmp_ln45_1 setgt auto
select_ln45_1_fu_424_p3 auto_sel 0	select_ln45_1 select
icmp_ln45_2_fu_436_p2 0	icmp_ln45_2 setgt auto
select_ln45_2_fu_442_p3 auto_sel 0	select_ln45_2 select
icmp_ln45_3_fu_454_p2 0	icmp_ln45_3 setgt auto
select_ln45_3_fu_460_p3 auto_sel 0	select_ln45_3 select
icmp_ln45_4_fu_472_p2 0	icmp_ln45_4 setgt auto
select_ln45_4_fu_478_p3 auto_sel 0	select_ln45_4 select
icmp_ln45_5_fu_490_p2 0	icmp_ln45_5 setgt auto
select_ln45_5_fu_496_p3 auto_sel 0	select_ln45_5 select
icmp_ln45_6_fu_508_p2 0	icmp_ln45_6 setgt auto

select_ln45_6_fu_514_p3	select_ln45_6 select
auto_sel 0	
icmp_ln45_7_fu_526_p2	icmp_ln45_7 setgt auto
0	
select_ln45_7_fu_532_p3	select_ln45_7 select
auto_sel 0	
icmp_ln45_8_fu_544_p2	icmp_ln45_8 setgt auto
0	
select_ln45_8_fu_550_p3	select_ln45_8 select
auto_sel 0	
icmp_ln45_9_fu_562_p2	icmp_ln45_9 setgt auto
0	
select_ln45_9_fu_568_p3	select_ln45_9 select
auto_sel 0	
icmp_ln45_10_fu_580_p2	icmp_ln45_10 setgt
auto 0	
select_ln45_10_fu_586_p3	select_ln45_10 select
auto_sel 0	
icmp_ln45_11_fu_598_p2	icmp_ln45_11 setgt auto
0	
select_ln45_11_fu_604_p3	select_ln45_11 select
auto_sel 0	
icmp_ln45_12_fu_616_p2	icmp_ln45_12 setgt
auto 0	
select_ln45_12_fu_622_p3	select_ln45_12 select
auto_sel 0	
icmp_ln45_13_fu_634_p2	icmp_ln45_13 setgt
auto 0	
select_ln45_13_fu_640_p3	select_ln45_13 select
auto_sel 0	
icmp_ln45_14_fu_652_p2	icmp_ln45_14 setgt
auto 0	
select_ln45_14_fu_658_p3	select_ln45_14 select
auto_sel 0	
icmp_ln45_15_fu_670_p2	icmp_ln45_15 setgt
auto 0	
select_ln45_15_fu_676_p3	select_ln45_15 select
auto_sel 0	
icmp_ln45_16_fu_688_p2	icmp_ln45_16 setgt
auto 0	
select_ln45_16_fu_694_p3	select_ln45_16 select
auto_sel 0	
icmp_ln45_17_fu_706_p2	icmp_ln45_17 setgt
auto 0	

select_ln45_17_fu_712_p3		select_ln45_17 select
auto_sel 0		
icmp_ln45_18_fu_724_p2		icmp_ln45_18 setgt
auto 0		
select_ln45_18_fu_730_p3		select_ln45_18 select
auto_sel 0		
icmp_ln45_19_fu_742_p2		icmp_ln45_19 setgt
auto 0		
select_ln45_19_fu_748_p3		select_ln45_19 select
auto_sel 0		
icmp_ln45_20_fu_760_p2		icmp_ln45_20 setgt
auto 0		
select_ln45_20_fu_766_p3		select_ln45_20 select
auto_sel 0		
icmp_ln45_21_fu_778_p2		icmp_ln45_21 setgt
auto 0		
select_ln45_21_fu_784_p3		select_ln45_21 select
auto_sel 0		
icmp_ln45_22_fu_796_p2		icmp_ln45_22 setgt
auto 0		
select_ln45_22_fu_802_p3		select_ln45_22 select
auto_sel 0		
icmp_ln45_23_fu_814_p2		icmp_ln45_23 setgt
auto 0		
select_ln45_23_fu_820_p3		select_ln45_23 select
auto_sel 0		
icmp_ln45_24_fu_832_p2		icmp_ln45_24 setgt
auto 0		
select_ln45_24_fu_838_p3		select_ln45_24 select
auto_sel 0		
icmp_ln45_25_fu_850_p2		icmp_ln45_25 setgt
auto 0		
select_ln45_25_fu_856_p3		select_ln45_25 select
auto_sel 0		
icmp_ln45_26_fu_868_p2		icmp_ln45_26 setgt
auto 0		
select_ln45_26_fu_874_p3		select_ln45_26 select
auto_sel 0		
icmp_ln45_27_fu_886_p2		icmp_ln45_27 setgt
auto 0		
select_ln45_27_fu_892_p3		select_ln45_27 select
auto_sel 0		
icmp_ln45_28_fu_904_p2		icmp_ln45_28 setgt
auto 0		

select_ln45_28_fu_910_p3	select_ln45_28 select
auto_sel 0	
icmp_ln45_29_fu_922_p2	icmp_ln45_29 setgt
auto 0	
select_ln45_29_fu_928_p3	select_ln45_29 select
auto_sel 0	
icmp_ln45_30_fu_940_p2	icmp_ln45_30 setgt
auto 0	
select_ln45_30_fu_946_p3	select_ln45_30 select
auto_sel 0	
icmp_ln45_31_fu_958_p2	icmp_ln45_31 setgt
auto 0	
select_ln45_31_fu_964_p3	select_ln45_31 select
auto_sel 0	
+ dense_latency_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_config4_s	30
mul_16s_12s_26_1_1_U336	1 mul_ln42 mul
auto 0	
mul_16s_11s_26_1_1_U334	1 mul_ln42_1 mul
auto 0	
mul_16s_11s_26_1_1_U351	1 mul_ln42_2 mul
auto 0	
mul_16s_9s_25_1_1_U337	1 mul_ln73 mul auto
0	
mul_16s_10ns_26_1_1_U328	1 mul_ln73_1 mul
auto 0	
mul_16s_11ns_26_1_1_U341	1 mul_ln42_3 mul
auto 0	
mul_16s_12ns_26_1_1_U340	1 mul_ln42_4 mul
auto 0	
mul_16s_11s_26_1_1_U348	1 mul_ln42_5 mul
auto 0	
sub_ln73_fu_4595_p2	sub_ln73 sub fabric
0	
mul_16s_12s_26_1_1_U326	1 mul_ln42_6 mul
auto 0	
mul_16s_11ns_26_1_1_U333	1 mul_ln42_7 mul
auto 0	
mul_16s_12ns_26_1_1_U346	1 mul_ln42_8 mul
auto 0	
mul_16s_12s_26_1_1_U325	1 mul_ln42_9 mul
auto 0	
mul_16s_12ns_26_1_1_U324	1 mul_ln42_10 mul
auto 0	

mul_16s_11s_26_1_1_U330 auto 0	1 mul_ln42_11 mul
mul_16s_10ns_26_1_1_U343 auto 0	1 mul_ln73_2 mul
mul_16s_10ns_26_1_1_U350 auto 0	1 mul_ln73_3 mul
mul_16s_11s_26_1_1_U347 auto 0	1 mul_ln42_12 mul
mul_16s_11s_26_1_1_U353 auto 0	1 mul_ln42_13 mul
mul_16s_10s_26_1_1_U332 auto 0	1 mul_ln73_4 mul
mul_16s_10ns_26_1_1_U338 auto 0	1 mul_ln73_5 mul
mul_16s_10s_26_1_1_U331 auto 0	1 mul_ln73_6 mul
sub_ln73_1_fu_4822_p2 0	sub_ln73_1 sub fabric
mul_16s_10s_26_1_1_U342 auto 0	1 mul_ln73_7 mul
mul_16s_10s_26_1_1_U344 auto 0	1 mul_ln73_8 mul
mul_16s_10s_26_1_1_U329 auto 0	1 mul_ln73_9 mul
mul_16s_11ns_26_1_1_U335 auto 0	1 mul_ln42_14 mul
mul_16s_10ns_26_1_1_U349 auto 0	1 mul_ln73_10 mul
mul_16s_11ns_26_1_1_U352 auto 0	1 mul_ln42_15 mul
mul_16s_11s_26_1_1_U327 auto 0	1 mul_ln42_16 mul
mul_16s_9s_25_1_1_U339 auto 0	1 mul_ln73_11 mul
mul_16s_11ns_26_1_1_U345 auto 0	1 mul_ln42_17 mul
add_ln58_fu_4981_p2 0	add_ln58 add fabric
add_ln58_1_fu_4987_p2 0	add_ln58_1 add fabric
add_ln58_4_fu_4999_p2 0	add_ln58_4 add fabric
add_ln58_7_fu_5011_p2 0	add_ln58_7 add fabric


```

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| + myproject | | | 1 | 0 | | | | |
|
| + sigmoid_ap_fixed_16_6_5_3_0_ap_fixed_16_6_5_3_0_sigmoid_config5_s | | | 1 |
0 | | | | | | |
| sigmoid_table_U | rom_1p | | 1 | | | sigmoid_table
| auto | 1 | 10, 1024, 1 |
+-----+-----+-----+-----+-----+
---+---+---+---+---+

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=====
== Pragma Report
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```

* Ignored Pragas

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```

+-----+-----+-----+-----+-----+
---+---+---+---+---+
| Type    | Options | Location |
| Messages | | |
+-----+-----+-----+-----+
---+---+---+---+---+
| ALLOCATION | instances = softmax<CONFIG_T> limit = CONFIG_T::parallelization_factor
function | in softmax_multidim (firmware/nnet_utils/nnet_activation.h:402) | unexpected pragma
argument 'softmax', expects function/operation |
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```

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* Pragas with Warnings

```

```

+-----+-----+-----+-----+-----+
---+---+---+---+---+
| Type    | Options | Location |
| Messages | | |
+-----+-----+-----+-----+
---+---+---+---+---+
| RESOURCE | variable=weights core=ROM_nP_BRAM | in dense_resource_rf_leq_nin
(firmware/nnet_utils/nnet_dense_resource.h:33) | 'Resource pragma' is deprecated, use
'bind_op/bind_storage pragma' instead |
| RESOURCE | variable=weights core=ROM_nP_BRAM | in dense_resource_rf_gt_nin_rem0
(firmware/nnet_utils/nnet_dense_resource.h:107) | 'Resource pragma' is deprecated, use
'bind_op/bind_storage pragma' instead |
| RESOURCE | variable=weights core=ROM_nP_BRAM | in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:189) | 'Resource pragma' is deprecated, use
'bind_op/bind_storage pragma' instead |

```

```
+-----+-----+-----+
+-----+-----+-----+
```

* Valid Pragma Syntax

```
+-----+-----+-----+
+-----+-----+-----+
| Type          | Options                                | Location
|
+-----+-----+-----+
```

```
+-----+-----+-----+
| ARRAY_RESHAPE  | variable=input_1 complete dim=0      | in
myproject (firmware/myproject.cpp:13) |
| ARRAY_PARTITION | variable=layer5_out complete dim=0   | in
myproject (firmware/myproject.cpp:14) |
| INTERFACE      | ap_vld port=input_1,layer5_out        | in myproject
(firmware/myproject.cpp:15)          |
| PIPELINE       |                                         | in myproject
(firmware/myproject.cpp:16)          |
| ARRAY_PARTITION | variable=layer2_out complete dim=0    | in
myproject (firmware/myproject.cpp:35) |
| ARRAY_PARTITION | variable=layer3_out complete dim=0    | in
myproject (firmware/myproject.cpp:38) |
| ARRAY_PARTITION | variable=layer4_out complete dim=0    | in
myproject (firmware/myproject.cpp:41) |
| PIPELINE       |                                         | in linear
(firmware/nnet_utils/nnet_activation.h:29) |
| PIPELINE       |                                         | in relu
(firmware/nnet_utils/nnet_activation.h:40) |
| PIPELINE       |                                         | in relu_max
(firmware/nnet_utils/nnet_activation.h:54) |
| PIPELINE       |                                         | in sigmoid
(firmware/nnet_utils/nnet_activation.h:109) |
| PIPELINE       |                                         | in softmax_latency
(firmware/nnet_utils/nnet_activation.h:176) |
| ARRAY_PARTITION | variable=exp_res complete              | in
softmax_latency (firmware/nnet_utils/nnet_activation.h:198) |
| UNROLL         |                                         | in softmax_latency
(firmware/nnet_utils/nnet_activation.h:201) |
| UNROLL         |                                         | in softmax_latency
(firmware/nnet_utils/nnet_activation.h:214) |
| PIPELINE       |                                         | in softmax_stable
(firmware/nnet_utils/nnet_activation.h:221) |
| UNROLL         |                                         | in softmax_stable
(firmware/nnet_utils/nnet_activation.h:247) |
```

ARRAY_PARTITION	variable=exp_res complete	in
softmax_stable (firmware/nnet_utils/nnet_activation.h:253)		
UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation.h:256)		
UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation.h:269)		
PIPELINE		in softmax_legacy
(firmware/nnet_utils/nnet_activation.h:317)		
UNROLL		in softmax_argmax
(firmware/nnet_utils/nnet_activation.h:362)		
PIPELINE		in softmax_argmax
(firmware/nnet_utils/nnet_activation.h:370)		
INLINE		in softmax
(firmware/nnet_utils/nnet_activation.h:382)		
INLINE		in softmax_multidim
(firmware/nnet_utils/nnet_activation.h:401)		
UNROLL		in softmax_multidim
(firmware/nnet_utils/nnet_activation.h:406)		
UNROLL		in softmax_multidim
(firmware/nnet_utils/nnet_activation.h:408)		
UNROLL		in softmax_multidim
(firmware/nnet_utils/nnet_activation.h:410)		
UNROLL		in softmax_multidim
(firmware/nnet_utils/nnet_activation.h:415)		
PIPELINE		in tanh
(firmware/nnet_utils/nnet_activation.h:452)		
FUNCTION_INSTANTIATE variable=table		in unary_lut
(firmware/nnet_utils/nnet_activation.h:481)		
ARRAY_PARTITION variable=table		in unary_lut
(firmware/nnet_utils/nnet_activation.h:482)		
UNROLL		in unary_lut
(firmware/nnet_utils/nnet_activation.h:485)		
PIPELINE		in hard_sigmoid
(firmware/nnet_utils/nnet_activation.h:496)		
PIPELINE		in hard_tanh
(firmware/nnet_utils/nnet_activation.h:511)		
PIPELINE		in leaky_relu
(firmware/nnet_utils/nnet_activation.h:529)		
PIPELINE		in thresholded_relu
(firmware/nnet_utils/nnet_activation.h:546)		
PIPELINE		in softplus
(firmware/nnet_utils/nnet_activation.h:591)		
PIPELINE		in softsign
(firmware/nnet_utils/nnet_activation.h:640)		

PIPELINE		in elu
(firmware/nnet_utils/nnet_activation.h:689)		
PIPELINE		in selu
(firmware/nnet_utils/nnet_activation.h:748)		
PIPELINE		in prelu
(firmware/nnet_utils/nnet_activation.h:776)		
PIPELINE		in binary_tanh
(firmware/nnet_utils/nnet_activation.h:804)		
PIPELINE		in ternary_tanh
(firmware/nnet_utils/nnet_activation.h:824)		
PIPELINE		in linear
(firmware/nnet_utils/nnet_activation_stream.h:20)		
UNROLL		in linear
(firmware/nnet_utils/nnet_activation_stream.h:28)		
PIPELINE		in relu
(firmware/nnet_utils/nnet_activation_stream.h:42)		
UNROLL		in relu
(firmware/nnet_utils/nnet_activation_stream.h:50)		
PIPELINE		in sigmoid
(firmware/nnet_utils/nnet_activation_stream.h:81)		
UNROLL		in sigmoid
(firmware/nnet_utils/nnet_activation_stream.h:89)		
ARRAY_PARTITION	variable=exp_res complete	in
softmax_latency (firmware/nnet_utils/nnet_activation_stream.h:133)		
PIPELINE	ii=ii	in softmax_latency
(firmware/nnet_utils/nnet_activation_stream.h:137)		
UNROLL		in softmax_latency
(firmware/nnet_utils/nnet_activation_stream.h:142)		
UNROLL		in softmax_latency
(firmware/nnet_utils/nnet_activation_stream.h:160)		
ALLOCATION	operation instances=mul limit=multiplier_limit	in
softmax_latency (firmware/nnet_utils/nnet_activation_stream.h:161)		
ARRAY_PARTITION	variable=data_array complete	in
softmax_stable (firmware/nnet_utils/nnet_activation_stream.h:193)		
PIPELINE	ii=ii	in softmax_stable
(firmware/nnet_utils/nnet_activation_stream.h:196)		
UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation_stream.h:201)		
UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation_stream.h:212)		
ARRAY_PARTITION	variable=exp_res complete	in
softmax_stable (firmware/nnet_utils/nnet_activation_stream.h:218)		
UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation_stream.h:221)		

UNROLL		in softmax_stable
(firmware/nnet_utils/nnet_activation_stream.h:239)		
ALLOCATION	operation instances=mul limit=multiplier_limit	in
softmax_stable (firmware/nnet_utils/nnet_activation_stream.h:240)		
PIPELINE		in softmax_legacy
(firmware/nnet_utils/nnet_activation_stream.h:272)		
UNROLL		in softmax_legacy
(firmware/nnet_utils/nnet_activation_stream.h:276)		
UNROLL		in softmax_legacy
(firmware/nnet_utils/nnet_activation_stream.h:283)		
UNROLL		in softmax_legacy
(firmware/nnet_utils/nnet_activation_stream.h:286)		
UNROLL		in softmax_legacy
(firmware/nnet_utils/nnet_activation_stream.h:309)		
PIPELINE		in softmax_argmax
(firmware/nnet_utils/nnet_activation_stream.h:326)		
UNROLL		in softmax_argmax
(firmware/nnet_utils/nnet_activation_stream.h:331)		
PIPELINE		in softmax_argmax
(firmware/nnet_utils/nnet_activation_stream.h:339)		
PIPELINE		in tanh
(firmware/nnet_utils/nnet_activation_stream.h:390)		
UNROLL		in tanh
(firmware/nnet_utils/nnet_activation_stream.h:398)		
FUNCTION_INSTANTIATE variable=table		in unary_lut
(firmware/nnet_utils/nnet_activation_stream.h:418)		
ARRAY_PARTITION variable=table complete		in unary_lut
(firmware/nnet_utils/nnet_activation_stream.h:419)		
PIPELINE ii=CONFIG_T::reuse_factor rewind		in unary_lut
(firmware/nnet_utils/nnet_activation_stream.h:423)		
UNROLL		in unary_lut
(firmware/nnet_utils/nnet_activation_stream.h:431)		
PIPELINE		in hard_sigmoid
(firmware/nnet_utils/nnet_activation_stream.h:449)		
UNROLL		in hard_sigmoid
(firmware/nnet_utils/nnet_activation_stream.h:457)		
PIPELINE		in hard_tanh
(firmware/nnet_utils/nnet_activation_stream.h:474)		
UNROLL		in hard_tanh
(firmware/nnet_utils/nnet_activation_stream.h:482)		
PIPELINE		in leaky_relu
(firmware/nnet_utils/nnet_activation_stream.h:503)		
UNROLL		in leaky_relu
(firmware/nnet_utils/nnet_activation_stream.h:511)		

PIPELINE		in thresholded_relu
(firmware/nnet_utils/nnet_activation_stream.h:529)		
UNROLL		in thresholded_relu
(firmware/nnet_utils/nnet_activation_stream.h:537)		
PIPELINE		in softplus
(firmware/nnet_utils/nnet_activation_stream.h:568)		
UNROLL		in softplus
(firmware/nnet_utils/nnet_activation_stream.h:576)		
PIPELINE		in softsign
(firmware/nnet_utils/nnet_activation_stream.h:609)		
UNROLL		in softsign
(firmware/nnet_utils/nnet_activation_stream.h:617)		
PIPELINE		in elu
(firmware/nnet_utils/nnet_activation_stream.h:650)		
UNROLL		in elu
(firmware/nnet_utils/nnet_activation_stream.h:658)		
PIPELINE		in selu
(firmware/nnet_utils/nnet_activation_stream.h:698)		
UNROLL		in selu
(firmware/nnet_utils/nnet_activation_stream.h:706)		
PIPELINE		in prelu
(firmware/nnet_utils/nnet_activation_stream.h:730)		
UNROLL		in prelu
(firmware/nnet_utils/nnet_activation_stream.h:738)		
PIPELINE		in binary_tanh
(firmware/nnet_utils/nnet_activation_stream.h:756)		
UNROLL		in binary_tanh
(firmware/nnet_utils/nnet_activation_stream.h:765)		
PIPELINE		in ternary_tanh
(firmware/nnet_utils/nnet_activation_stream.h:784)		
UNROLL		in ternary_tanh
(firmware/nnet_utils/nnet_activation_stream.h:792)		
ARRAY_PARTITION	variable=data_buf complete dim=0	in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:19)		
ARRAY_PARTITION	variable=mult complete	in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:22)		
ARRAY_PARTITION	variable=acc complete	in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:25)		
ARRAY_PARTITION	variable=weights complete	in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:27)		
ARRAY_PARTITION	variable=biases complete	in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:28)		
ALLOCATION	operation instances=mul limit=CONFIG_T::mult_config::multiplier_limit	
in conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:31)		

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| PIPELINE          | ii=CONFIG_T::reuse_factor rewind          | in
conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:35) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:41) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:48) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:52) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:62) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:69) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:72) |
| UNROLL           |          | in conv_1d_latency_cl
(firmware/nnet_utils/nnet_conv1d_latency.h:80) |
| ARRAY_PARTITION   | variable=mult complete dim=0              | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:98) |
| ARRAY_PARTITION   | variable=acc complete dim=0              | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:99) |
| FUNCTION_INSTANTIATE | variable=weights,biases                  | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:102) |
| PIPELINE          | ii=CONFIG_T::reuse_factor                | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:105) |
| ARRAY_PARTITION   | variable=weights complete dim=0          | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:106) |
| ARRAY_PARTITION   | variable=biases complete dim=0           | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:107) |
| ALLOCATION          | operation instances=mul limit=CONFIG_T::mult_config::multiplier_limit |
in pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:110) |
| UNROLL           |          | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:119) |
| UNROLL           |          | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:138) |
| UNROLL           |          | in
pointwise_conv_1d_latency_cl (firmware/nnet_utils/nnet_conv1d_latency.h:160) |
| INLINE           |          | in dense
(firmware/nnet_utils/nnet_dense.h:44) |
| INLINE           |          | in dense
(firmware/nnet_utils/nnet_dense.h:53) |
| INLINE           |          | in dense
(firmware/nnet_utils/nnet_dense.h:64) |
| INLINE           |          | in dense
(firmware/nnet_utils/nnet_dense.h:75) |

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| UNROLL | | in fill_mult
(firmware/nnet_utils/nnet_dense_compressed.h:15) |
| ARRAY_PARTITION | variable=acc complete | in
dense_compressed (firmware/nnet_utils/nnet_dense_compressed.h:29) |
| ARRAY_PARTITION | variable=biases complete | in
dense_compressed (firmware/nnet_utils/nnet_dense_compressed.h:30) |
| ARRAY_RESHAPE | variable=weights block factor=multiplier_limit | in
dense_compressed (firmware/nnet_utils/nnet_dense_compressed.h:31) |
| AGGREGATE | variable=weights | in
dense_compressed (firmware/nnet_utils/nnet_dense_compressed.h:34) |
| UNROLL | | in dense_compressed
(firmware/nnet_utils/nnet_dense_compressed.h:41) |
| PIPELINE | ii=1 rewind | in dense_compressed
(firmware/nnet_utils/nnet_dense_compressed.h:49) |
| ARRAY_PARTITION | variable=mult complete | in
dense_compressed (firmware/nnet_utils/nnet_dense_compressed.h:52) |
| UNROLL | | in dense_compressed
(firmware/nnet_utils/nnet_dense_compressed.h:56) |
| UNROLL | | in dense_compressed
(firmware/nnet_utils/nnet_dense_compressed.h:62) |
| UNROLL | | in dense_compressed
(firmware/nnet_utils/nnet_dense_compressed.h:82) |
| FUNCTION_INSTANTIATE | variable=weights,biases | in
dense_latency (firmware/nnet_utils/nnet_dense_latency.h:21) |
| PIPELINE | ii=CONFIG_T::reuse_factor | in dense_latency
(firmware/nnet_utils/nnet_dense_latency.h:26) |
| ARRAY_PARTITION | variable=biases complete | in
dense_latency (firmware/nnet_utils/nnet_dense_latency.h:29) |
| ARRAY_PARTITION | variable=mult complete | in
dense_latency (firmware/nnet_utils/nnet_dense_latency.h:30) |
| ARRAY_PARTITION | variable=acc complete | in
dense_latency (firmware/nnet_utils/nnet_dense_latency.h:31) |
| ALLOCATION | operation instances=mul limit=CONFIG_T::multiplier_limit | in
dense_latency (firmware/nnet_utils/nnet_dense_latency.h:33) |
| FUNCTION_INSTANTIATE | variable=weights,biases | in
dense_resource_rf_leq_nin (firmware/nnet_utils/nnet_dense_resource.h:28) |
| ARRAY_RESHAPE | variable=weights block factor=block_factor | in
dense_resource_rf_leq_nin (firmware/nnet_utils/nnet_dense_resource.h:29) |
| ARRAY_PARTITION | variable=biases complete | in
dense_resource_rf_leq_nin (firmware/nnet_utils/nnet_dense_resource.h:30) |
| ARRAY_PARTITION | variable=acc complete | in
dense_resource_rf_leq_nin (firmware/nnet_utils/nnet_dense_resource.h:37) |
| UNROLL | | in dense_resource_rf_leq_nin
(firmware/nnet_utils/nnet_dense_resource.h:41) |

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| PIPELINE          | ii=1 rewind                      | in
dense_resource_rf_leq_nin (firmware/nnet_utils/nnet_dense_resource.h:47) |
| UNROLL           |                                | in dense_resource_rf_leq_nin
(firmware/nnet_utils/nnet_dense_resource.h:56) |
| UNROLL           |                                | in dense_resource_rf_leq_nin
(firmware/nnet_utils/nnet_dense_resource.h:81) |
| FUNCTION_INSTANTIATE | variable=weights,biases          | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:102) |
| ARRAY_RESHAPE      | variable=weights block factor=block_factor          | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:103) |
| ARRAY_PARTITION    | variable=biases complete          | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:104) |
| ARRAY_PARTITION    | variable=acc complete             | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:111) |
| UNROLL             |                                | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:115) |
| PIPELINE          | ii=1 rewind                      | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:136) |
| UNROLL             |                                | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:143) |
| UNROLL             |                                | in
dense_resource_rf_gt_nin_rem0 (firmware/nnet_utils/nnet_dense_resource.h:163) |
| FUNCTION_INSTANTIATE | variable=weights,biases          | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:184) |
| ARRAY_RESHAPE      | variable=weights block factor=block_factor          | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:185) |
| ARRAY_PARTITION    | variable=biases complete          | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:186) |
| ARRAY_PARTITION    | variable=acc complete             | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:193) |
| UNROLL             |                                | in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:197) |
| PIPELINE          | ii=1 rewind                      | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:203) |
| ARRAY_PARTITION    | variable=tmpmult complete          | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:205) |
| UNROLL             |                                | in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:209) |
| ARRAY_PARTITION    | variable=mult complete             | in
dense_resource_rf_gt_nin (firmware/nnet_utils/nnet_dense_resource.h:219) |
| UNROLL             |                                | in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:223) |
| UNROLL             |                                | in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:229) |

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UNROLL		in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:239)		
UNROLL		in dense_resource_rf_gt_nin
(firmware/nnet_utils/nnet_dense_resource.h:249)		
INLINE	recursive	in dense_resource
(firmware/nnet_utils/nnet_dense_resource.h:259)		
PIPELINE	ii=CONFIG_T::reuse_factor	in
dense_latency_wrapper (firmware/nnet_utils/nnet_dense_stream.h:16)		
INLINE		in data_prepare
(firmware/nnet_utils/nnet_dense_stream.h:29)		
PIPELINE		in data_prepare
(firmware/nnet_utils/nnet_dense_stream.h:34)		
UNROLL		in data_prepare
(firmware/nnet_utils/nnet_dense_stream.h:38)		
UNROLL		in data_prepare
(firmware/nnet_utils/nnet_dense_stream.h:46)		
INLINE		in res_write
(firmware/nnet_utils/nnet_dense_stream.h:54)		
PIPELINE		in res_write
(firmware/nnet_utils/nnet_dense_stream.h:59)		
UNROLL		in res_write
(firmware/nnet_utils/nnet_dense_stream.h:64)		
UNROLL		in res_write
(firmware/nnet_utils/nnet_dense_stream.h:74)		
INLINE	recursive	in dense
(firmware/nnet_utils/nnet_dense_stream.h:85)		
ARRAY_PARTITION	variable=data complete	in dense
(firmware/nnet_utils/nnet_dense_stream.h:88)		
ARRAY_PARTITION	variable=res complete	in dense
(firmware/nnet_utils/nnet_dense_stream.h:91)		
INLINE		in bit_shift
(firmware/nnet_utils/nnet_function_stubs.h:59)		
INLINE		in bit_shift
(firmware/nnet_utils/nnet_function_stubs.h:66)		
INLINE		in bit_shift
(firmware/nnet_utils/nnet_function_stubs.h:73)		
INLINE		in bit_shift
(firmware/nnet_utils/nnet_function_stubs.h:80)		
INLINE		in product
(firmware/nnet_utils/nnet_mult.h:25)		
INLINE		in product
(firmware/nnet_utils/nnet_mult.h:34)		
INLINE		in product
(firmware/nnet_utils/nnet_mult.h:46)		

INLINE		in product
(firmware/nnet_utils/nnet_mult.h:58)		
INLINE		in product
(firmware/nnet_utils/nnet_mult.h:72)		
INLINE		in product
(firmware/nnet_utils/nnet_mult.h:82)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:23)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:33)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:47)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:59)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:76)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:90)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:109)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:125)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:146)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:164)		
PIPELINE		in clone_stream
(firmware/nnet_utils/nnet_stream.h:187)		
UNROLL		in clone_stream
(firmware/nnet_utils/nnet_stream.h:207)		
PIPELINE		in repack_stream
(firmware/nnet_utils/nnet_stream.h:230)		
UNROLL		in repack_stream
(firmware/nnet_utils/nnet_stream.h:237)		
PIPELINE		in repack_stream
(firmware/nnet_utils/nnet_stream.h:247)		
PIPELINE		in repack_stream
(firmware/nnet_utils/nnet_stream.h:255)		
UNROLL		in repack_stream
(firmware/nnet_utils/nnet_stream.h:259)		
PIPELINE		in repack_stream
(firmware/nnet_utils/nnet_stream.h:270)		
UNROLL		in repack_stream
(firmware/nnet_utils/nnet_stream.h:274)		

PIPELINE		in broadcast_stream_1x1xc
(firmware/nnet_utils/nnet_stream.h:295)		
PIPELINE		in broadcast_stream_1x1xc
(firmware/nnet_utils/nnet_stream.h:298)		
UNROLL		in broadcast_stream_1x1xc
(firmware/nnet_utils/nnet_stream.h:302)		
PIPELINE		in broadcast_stream_hxwx1
(firmware/nnet_utils/nnet_stream.h:316)		
UNROLL		in broadcast_stream_hxwx1
(firmware/nnet_utils/nnet_stream.h:321)		
UNROLL		in operator=
(firmware/nnet_utils/nnet_types.h:28)		
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Overview

Name
▼ myproject
dense_latency<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, config2>
relu<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, relu_config3>
dense_latency<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, config4>
sigmoid<ap_fixed<16, 6, 5, 3, 0>, ap_fixed<16, 6, 5, 3, 0>, sigmoid_config5>